

Linear chain

Assignment #3

1 The undistorted 1-D hydrogen chain

Calculate with VASP the ground state of the hydrogen one-dimensional (1-D) chain, considering identical interatomic distances between every atomic pair (*i.e.*, undistorted H chain).

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- « Build a supercell with vectors $(a,0,0)$, $(0,b,0)$ and $(0,0,b)$ (with $b \gg a$), including 2 atoms; place one atom in the origin $(0,0,0)$, and one atom in $(a/2,0,0)$.
- « Construct the KPOINTS file with a Monkhorst-Pack (MP, see KPOINTS page in VASP wiki) grid generated automatically, including only one k point in the $\vec{b}_2$ and $\vec{b}_3$ reciprocal-space directions; $\vec{b}_2$ and $\vec{b}_3$ correspond to the real-space directions perpendicular to the H chain (*i.e.*, y and z).
- « Calculate the energy of the system as a function of the lattice parameter a (*e.g.*, from 1.0 Å to 4.0 Å with steps of 0.25 Å), using a rough but reasonable number $\tilde{k}$ of k points along the $\vec{b}_1$ direction (*e.g.*, $\tilde{k} = 5$).
- « Determine the ground-state interatomic distance in the rough $\tilde{k} \times 1 \times 1$ setup.
- « Use the ground-state structure calculated in the rough setup to perform convergence test calculations on the k -point mesh: vary the number of k points along the $\vec{b}_1$ direction, up to reaching the convergence criteria on the total energy of 1 meV/atom per variation of 1 k point.
- » Include the convergence test in your report.
- « Re-calculate the ground state of the system, using the k -point setup which guarantees well converged results: set up the KPOINTS according to your convergence test and calculate the energy of the system as a function of the lattice parameter a (vary a in a small interval around the value previously calculated in the rough setup, *e.g.*, $\pm 10\%$).
- » Plot the energy as a function of the lattice parameter a and indicate the ground-state interatomic distance $a_0/2$ (by fitting the data with a polynomial expansion of the second order).

2 Peierls distortions in the 1-D hydrogen chain

Calculate with VASP the ground state of the hydrogen one-dimensional (1-D) chain, considering Peierls distortions, namely a dimerization of the Hydrogen atoms.

Compare the energies of the undistorted (from Ex. 1) and distorted chains and indicate which solution is more favorable. According to these results, can you make a prediction on the ground-state of the real system?

- « Use the optimized supercell and setup from Exercise 1, and vary the distance between the two H atoms in the unit cell, while keeping constant the cell size.
- » Plot the energy as a function of the deviation of the interatomic distances from the $a_0/2$ value calculated in Ex. 1.

3 Electronic energy-band structure

Calculate with VASP the electronic energy-band structures $E_n(k)$ of the undistorted and distorted 1-D hydrogen chains (in their respective ground states).

What is the difference between these two band structures? Can you explain the reason?

- « Perform a calculation on both systems in order to get self-consistent electron density and wavefunctions.
- « Perform non-self consistent calculations to get the electronic band structure (set ICHARG=11); avoid to write electron density and wavefunctions (LCHARG=.FALSE. and LWAVE=.FALSE.). Set LORBIT=10 (see vaspkit point below).
- « In the non-self consistent calculations, use the line mode in the KPOINTS file (see [Strings of k-points for bandstructure calculations](#): use a large number of k points along the $\vec{b}_1$ direction, in order to get a smooth $E_n(k)$ dispersion, between $(-b_1, 0, 0)$ and $(b_1, 0, 0)$.
- » For a quick, visual check of the calculated band structure, you can use the vaspkit tool (see instruction in basics.pdf). MIND: vaspkit requires the output file PROCAR, which is created by setting LORBIT=10 in the INCAR file.
- » Extract the $E_n(k)$ data from the OUTCAR or EIGENVAL file, by using a script or vaspkit
- » Plot the energies of the lowest two bands ($n = 1, 2$) as a function of the k -point along $\vec{b}_1$ (bandstructure). Graphically distinguish filled and empty states by using different colors, or line width, or line style. Mark the Brillouin zone centered around the $\Gamma = (0, 0, 0)$ point.

4 (Optional) From 1D to 2D: graphene

General considerations:

- « Use GGA pseudo-potentials.
- « In the INCAR file set ISMEAR=-5 for self consistent calculations, and ISMEAR=0 and SIGMA=0.01 for band structure calculations; perform spin un-polarized calcu-

lations.

- « Construct the films on the xy plane (*i.e.*, perpendicularly to the z direction); set $\vec{a}_3 = (0, 0, 15 \text{ \AA})$. Mind that $\vec{a}_3$ depends on the lattice constant prefactor a_0 .
- « In serial sets of calculations (such as the k -mesh convergence tests, or calculations to determine the lattice constant) enable writing and reading of the wavefunctions in order to speed up the calculations.

Calculate with VASP the ground state of graphene (hexagonal C monolayer), by varying the lattice constant prefactor a_0 . Compare the result obtained by density functional theory with the experimental data (experimental lattice vector length $a^{\text{exp}} = 2.455 \text{ \AA}$).

Study the effect of the k mesh on the DOS, by considering only a Gamma centered grid and by varying the number of k points. In particular, include the energy band gap in your analysis. How many irreducible k points are required in order to obtain a converged DOS?

Calculate and discuss the band structure, in particular the eigenvalues around the Fermi level.

- « Build a primitive cell including two atoms of C.
- « Construct the KPOINTS file by defining a $7 \times 7 \times 1$ grid centered at the $\Gamma = (0, 0, 0)$ point.
- « Calculate the energy of the system as a function of the supercell size by varying the lattice constant prefactor a_0 (*e.g.*, from $2.30 \text{ \AA}$ to $2.60 \text{ \AA}$ with steps of $0.01 \text{ \AA}$).
- » Determine the value of a_0 yielding to the ground state, by fitting the calculated data.
- « Use the ground-state structure to analyze the effect of the k mesh on the DOS: use Gamma centered k -point grids; vary the number of k points along the $\vec{b}_1$ and $\vec{b}_2$ directions, simultaneously (*i.e.*, set $k_i \times k_i \times 1$ with k_i going from 5 to 20); compare the DOS obtained with the various setups, commenting on the energy gap.
- « Calculate the electronic energy-band structure: firstly, perform a self-consistent calculation, writing both the WAVECAR and CHGCAR files, by using an accurate k -point grid; finally, perform a non-self-consistent calculation, by reading the wavefunctions and charge density (ICHARG=11); use the K'- Γ -M-K-K' path for the band structure.